

EXTENSION 2

Complex Numbers (Ext2), N1 Introduction to Complex Numbers (Ext2)

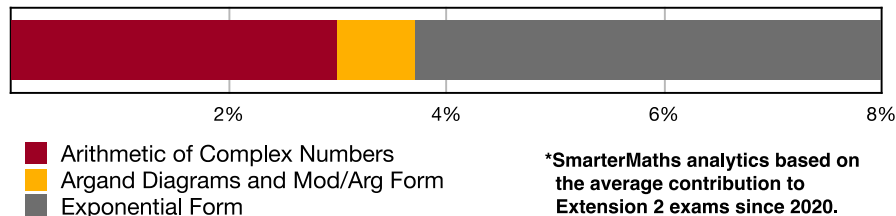
Argand Diagrams and Mod/Arg form

Teacher: Nardene Dodd

Exam Equivalent Time: 102 minutes (based on allocation of 1.5 minutes per mark)



N1 Introduction to Complex Numbers



HISTORICAL CONTRIBUTION

- *N1 Introduction to Complex Numbers* has contributed 8.0% per new syllabus Ext2 exam since it was introduced in 2020.
- We have split this topic into three categories for analysis purposes which are: 1-Arithmetic of Complex Numbers (3.0%), 2-Argand Diagrams and Mod/Arg Form (0.7%) and 3-Exponential Form (4.3%).
- This analysis looks at *Argand Diagrams and Mod/Arg Form*.

HSC ANALYSIS - What to expect and common pitfalls

- *Argand Diagrams and Mod/Arg Form* (0.7%) was regularly examined pre-2020 in 4-5 mark questions although we note this has fallen off a cliff with just a single dedicated multiple choice in the first three Ext2 exams combined.
- Although dedicated questions with significant mark allocations look unlikely at this stage to return, this topic area represents important foundational knowledge for more difficult later topics within N2 *Using Complex Numbers*.
- The most common question type is converting the Cartesian form of a complex number into mod/arg form and then raising it to a power using De Moivre. The database includes numerous examples of this.
- Revision should cover all basic identities involving modulus and argument. Note 2020 Ext2 4 MC where one third of students answered incorrectly.
- Revision of 2008 Ext2 N1 8b is recommended where equating real parts of two answers is used to find the exact surd form of a trig ratio.

Questions

1. Complex Numbers, EXT2 N1 2014 HSC 4 MC

Given $z = 2\left(\cos \frac{\pi}{3} + i\sin \frac{\pi}{3}\right)$, which expression is equal to $(\bar{z})^{-1}$?

- (A) $\frac{1}{2}\left(\cos \frac{\pi}{3} - i\sin \frac{\pi}{3}\right)$
- (B) $2\left(\cos \frac{\pi}{3} - i\sin \frac{\pi}{3}\right)$
- (C) $\frac{1}{2}\left(\cos \frac{\pi}{3} + i\sin \frac{\pi}{3}\right)$
- (D) $2\left(\cos \frac{\pi}{3} + i\sin \frac{\pi}{3}\right)$

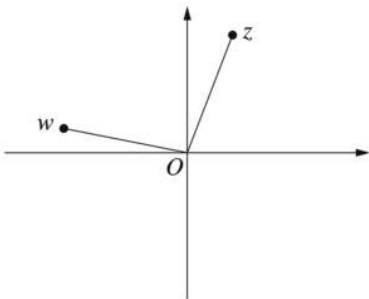
2. Complex Numbers, EXT2 N1 2015 HSC 5 MC

Given that $z = 1 - i$, which expression is equal to z^3 ?

- (A) $\sqrt{2}\left(\cos\left(\frac{-3\pi}{4}\right) + i\sin\left(\frac{-3\pi}{4}\right)\right)$
- (B) $2\sqrt{2}\left(\cos\left(\frac{-3\pi}{4}\right) + i\sin\left(\frac{-3\pi}{4}\right)\right)$
- (C) $\sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$
- (D) $2\sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$

3. Complex Numbers, EXT2 N1 2016 HSC 4 MC

The Argand diagram shows the complex numbers z and w , where z lies in the first quadrant and w lies in the second quadrant.

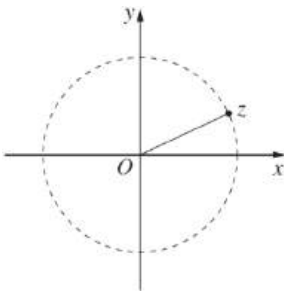


Which complex number could lie in the 3rd quadrant?

- (A) $-w$
 - (B) $2iz$
 - (C) $\bar{z}$
 - (D) $w - z$
-

4. Complex Numbers, EXT2 N1 2012 HSC 3 MC

The complex number z is shown on the Argand diagram below.



Which of the following best represents $i\bar{z}$?

- (A)

An Argand diagram with a dashed unit circle centered at origin O. A point labeled $i\bar{z}$ is in the first quadrant, connected to O by a line segment.
- (B)

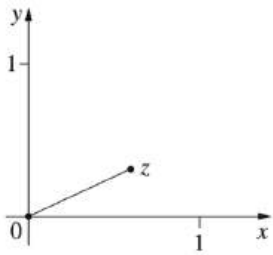
An Argand diagram with a dashed unit circle centered at origin O. A point labeled $i\bar{z}$ is in the second quadrant, connected to O by a line segment.
- (C)

An Argand diagram with a dashed unit circle centered at origin O. A point labeled $i\bar{z}$ is in the third quadrant, connected to O by a line segment.
- (D)

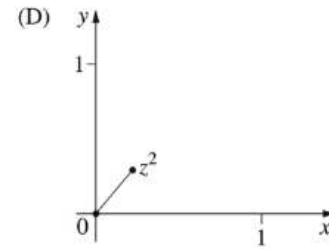
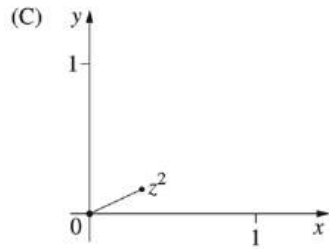
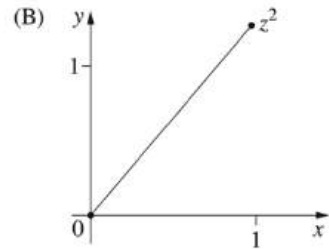
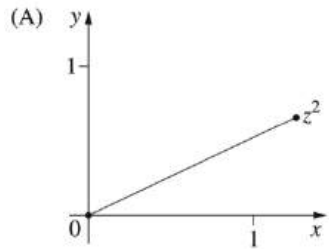
An Argand diagram with a dashed unit circle centered at origin O. A point labeled $i\bar{z}$ is in the fourth quadrant, connected to O by a line segment.
-

5. Complex Numbers, EXT2 N1 2013 HSC 3 MC

The Argand diagram below shows the complex number z .

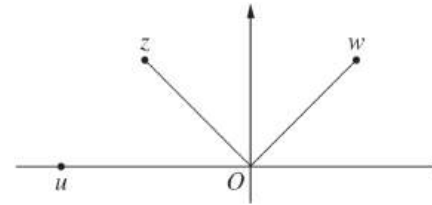


Which diagram best represents z^2 ?



6. Complex Numbers, EXT2 N1 2014 HSC 8 MC

The Argand diagram shows the complex numbers w, z and u , where w lies in the first quadrant, z lies in the second quadrant and u lies on the negative real axis.

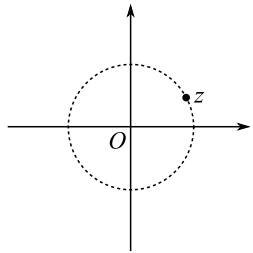


Which statement could be true?

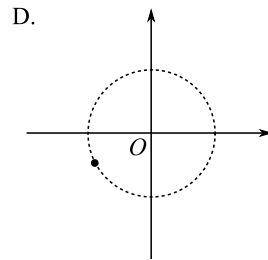
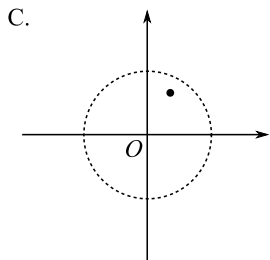
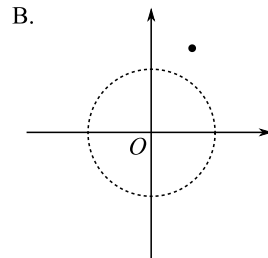
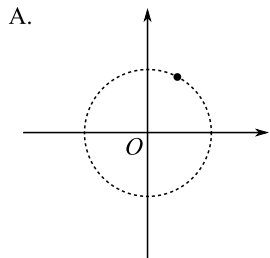
- (A) $u = zw$ and $u = z + w$
- (B) $u = zw$ and $u = z - w$
- (C) $z = uw$ and $u = z + w$
- (D) $z = uw$ and $u = z - w$

7. Complex Numbers, EXT2 N1 2020 HSC 4 MC

The diagram shows the complex number z on the Argand diagram.



Which of the following diagrams best shows the position of $\frac{z^2}{|z|}$?



8. Complex Numbers, EXT2 N1 2021 SPEC2 4 MC

For the complex number z , if $\text{Im}(z) > 0$, then $\text{Arg}\left(\frac{z\bar{z}}{z - \bar{z}}\right)$ is

- A. $-\frac{\pi}{2}$
- B. 0
- C. $\frac{\pi}{4}$
- D. π

9. Complex Numbers, EXT2 N1 2012 HSC 11d

- i. Write $z = \sqrt{3} - i$ in modulus-argument form. (2 marks)
- ii. Hence express z^9 in the form $x + iy$, where x and y are real. (1 mark)

10. Complex Numbers, EXT2 N1 2010 HSC 2b

- i. Express $-\sqrt{3} - i$ in modulus-argument form. (2 marks)
- ii. Show that $(-\sqrt{3} - i)^6$ is a real number. (2 marks)

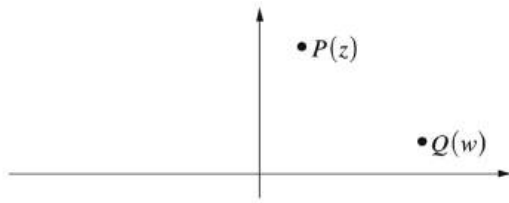
11. Complex Numbers, EXT2 N1 2016 HSC 11a

Let $z = \sqrt{3} - i$.

- i. Express z in modulus-argument form. (2 marks)
- ii. Show that z^6 is real. (1 mark)
- iii. Find a positive integer n such that z^n is purely imaginary. (1 mark)

12. Complex Numbers, EXT2 N1 2009 HSC 2c

The points P and Q on the Argand diagram represent the complex numbers z and w respectively.



Copy the diagram into your writing booklet, and mark on it the following points:

- the point R representing iz . (1 mark)
- the point S representing $\bar{w}$. (1 mark)
- the point T representing $z + w$. (1 mark)

13. Complex Numbers, EXT2 N1 2014 HSC 11a

Consider the complex numbers $z = -2 - 2i$ and $w = 3 + i$.

- Express $z + w$ in modulus-argument form. (2 marks)
- Express $\frac{z}{w}$ in the form $x + iy$, where x and y are real numbers. (2 marks)

14. Complex Numbers, EXT2 N1 2005 HSC 2b

Let $\beta = 1 - i\sqrt{3}$.

- Express β in modulus-argument form. (2 marks)
- Express β^5 in modulus-argument form. (2 marks)
- Hence express β^5 in the form $x + iy$. (1 mark)

15. Complex Numbers, EXT2 N1 2019 HSC 11e

Let $z = -1 + i\sqrt{3}$.

- Write z in modulus-argument form. (2 marks)
- Find z^3 , giving your answer in the form $x + iy$, where x and y are real numbers. (2 marks)

16. Complex Numbers, EXT2 N1 2004 HSC 2b

Let $\alpha = 1 + i\sqrt{3}$ and $\beta = 1 + i$.

- Find $\frac{\alpha}{\beta}$, in the form $x + iy$. (1 mark)
- Express α in modulus-argument form. (3 marks)
- Given that β has the modulus-argument form

$$\beta = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

find the modulus-argument form of $\frac{\alpha}{\beta}$. (1 mark)

- Hence find the exact value of $\sin \frac{\pi}{12}$. (1 mark)

17. Complex Numbers, EXT2 N1 2006 HSC 2b

- Express $\sqrt{3} - i$ in modulus-argument form. (2 marks)
- Express $(\sqrt{3} - i)^7$ in modulus-argument form. (2 marks)
- Hence express $(\sqrt{3} - i)^7$ in the form $x + iy$. (1 mark)

18. Complex Numbers, EXT2 N1 2007 HSC 2b

- Write $1 + i$ in the form $r(\cos \theta + i \sin \theta)$. (2 marks)
- Hence, or otherwise, find $(1 + i)^{17}$ in the form $a + ib$, where a and b are integers. (3 marks)

19. Complex Numbers, EXT2 N1 2018 HSC 13b

Let $z = 1 - \cos 2\theta + i \sin 2\theta$ where $0 < \theta \leq \pi$.

- Show that $|z| = 2 \sin \theta$. (2 marks)
- Show that $\arg(z) = \frac{\pi}{2} - \theta$. (2 marks)

20. Complex Numbers, EXT2 N1 2015 HSC 11b

Consider the complex numbers $z = -\sqrt{3} + i$ and $w = 3 \left(\cos \frac{\pi}{7} + i \sin \frac{\pi}{7} \right)$.

- Evaluate $|z|$. (1 mark)
- Evaluate $\arg(z)$. (1 mark)
- Find the argument of $\frac{z}{w}$. (1 mark)

21. Complex Numbers, EXT2 N1 2015 HSC 12a

The complex number z is such that $|z| = 2$ and $\arg(z) = \frac{\pi}{4}$.

Plot each of the following complex numbers on the same half-page Argand diagram.

- z (1 mark)
- $u = z^2$ (1 mark)
- $v = z^2 - \bar{z}$ (1 mark)

22. Complex Numbers, EXT2 N1 2008 HSC 2b

- Write $\frac{1+i\sqrt{3}}{1+i}$ in the form $x+iy$, where x and y are real. (2 marks)
- By expressing both $1+i\sqrt{3}$ and $1+i$ in modulus-argument form, write $\frac{1+i\sqrt{3}}{1+i}$ in modulus-argument form. (3 marks)
- Hence find $\cos\frac{\pi}{12}$ in surd form. (1 mark)
- By using the result of part (ii), or otherwise, calculate $\left(\frac{1+i\sqrt{3}}{1+i}\right)^{12}$. (1 mark)

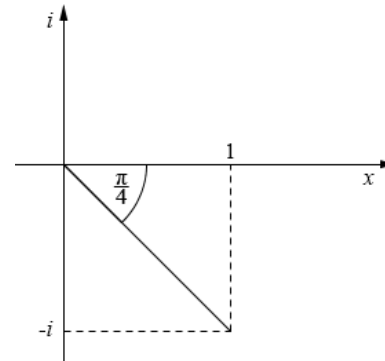
Worked Solutions

1. Complex Numbers, EXT2 N1 2014 HSC 4 MC

$$\begin{aligned} z &= 2\text{cis}\left(\frac{\pi}{3}\right) \\ \bar{z} &= 2\text{cis}\left(-\frac{\pi}{3}\right) \\ (\bar{z})^{-1} &= \left(2\text{cis}\left(-\frac{\pi}{3}\right)\right)^{-1} \\ &= 2^{-1}\text{cis}\left(\frac{\pi}{3}\right) \\ &= \frac{1}{2}\text{cis}\left(\frac{\pi}{3}\right) \\ &= \frac{1}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \end{aligned}$$

$\Rightarrow C$

2. Complex Numbers, EXT2 N1 2015 HSC 5 MC

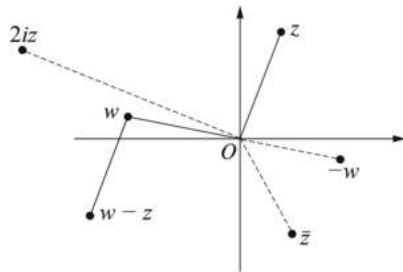


$$\begin{aligned} z &= 1 - i \\ |1 - i| &= \sqrt{2} \\ \arg(z) &= -\frac{\pi}{4} \\ z &= \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right) \\ \therefore z^3 &= 2\sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right) \quad (\text{De Moivre}) \end{aligned}$$

$\Rightarrow B$

3. Complex Numbers, EXT2 N1 2016 HSC 4 MC

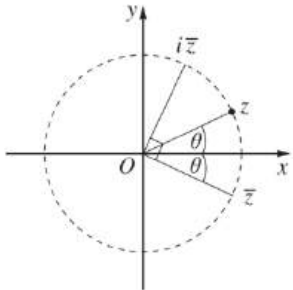
Using the parallelogram method:



From the graph above,
 $w-z$ could lie in 3rd quadrant.
 $\Rightarrow D$

4. Complex Numbers, EXT2 N1 2012 HSC 3 MC

$i\bar{z}$ is $\bar{z}$ rotated 90° anticlockwise.



$\Rightarrow A$

5. Complex Numbers, EXT2 N1 2013 HSC 3 MC

Consider z in polar form:

$$|z| < 1$$

$$\therefore |z^2| < |z|$$

$$\arg(z^2) = 2\arg(z)$$

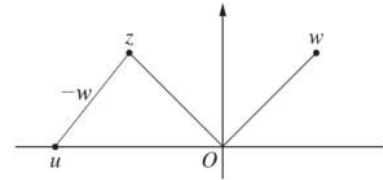
$\Rightarrow D$

6. Complex Numbers, EXT2 N1 2014 HSC 8 MC

Using the parallelogram method, it could be true that

$$u = z - w$$

$\Rightarrow$ Eliminate (A) and (C)



$$\arg(u) = \arg(w) + \arg(z)$$

$$u = zw$$

$\Rightarrow B$

7. Complex Numbers, EXT2 N1 2020 HSC 4 MC

Let $z = r \operatorname{cis} \theta$

$$z^2 = r^2 \operatorname{cis} (2\theta)$$

$$|z| = r$$

$$\therefore \frac{z^2}{|z|} = \frac{r^2 \operatorname{cis} (2\theta)}{r} = r \operatorname{cis} (2\theta)$$

On Argand diagram, it lies on the dotted line (modulus the same) with an argument that is doubled.

$\Rightarrow A$

8. Complex Numbers, EXT2 N1 2021 SPEC2 4 MC

Let $z = x + iy \Rightarrow \bar{z} = x - iy$

$$\begin{aligned} \operatorname{Arg}\left(\frac{z\bar{z}}{z-\bar{z}}\right) &= \operatorname{Arg}(z\bar{z}) - \operatorname{Arg}(z-\bar{z}) \\ &= \operatorname{Arg}(x^2 + y^2) - \operatorname{Arg}(2yi) \\ &= 0 - \operatorname{Arg}(2yi), \text{ where } y > 0 \\ &= -\frac{\pi}{2} \end{aligned}$$

$\Rightarrow A$

9. Complex Numbers, EXT2 N1 2012 HSC 11d

$$\text{i. } z = \sqrt{3} - i$$

$$|z| = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$\therefore z = \sqrt{3} - i = 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= 2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right)$$

$$= 2 \operatorname{cis} \left(-\frac{\pi}{6} \right)$$

$$\text{ii. } z^9 = 2^9 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right)^9$$

$$= 2^9 \operatorname{cis} \left(-\frac{9\pi}{6} \right) \quad (\text{by De Moivre})$$

$$= 512 \operatorname{cis} \left(-\frac{3\pi}{2} \right)$$

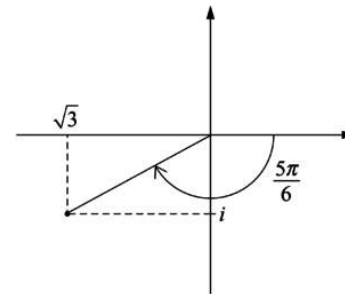
$$= 512(0 + i)$$

$$= i512$$

10. Complex Numbers, EXT2 N1 2010 HSC 2b

$$\text{i. } |-\sqrt{3} - i| = \sqrt{(-\sqrt{3})^2 + (-1)^2}$$

$$= 2$$



From the graph

$$\arg(-\sqrt{3} - i) = -\frac{5\pi}{6} \quad (\text{for } -\pi < \theta < \pi)$$

$$\therefore -\sqrt{3} - i = 2 \operatorname{cis} \left(-\frac{5\pi}{6} \right)$$

Alternative Solution (to find the argument)

$$-\sqrt{3} - i = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= 2 \operatorname{cis} \left(-\frac{5\pi}{6} \right)$$

$$\text{ii. } (-\sqrt{3} - i)^6 = \left[2 \operatorname{cis} \left(-\frac{5\pi}{6} \right) \right]^6$$

$$= 2^6 \left[\cos \left(\frac{-5\pi}{6} \times 6 \right) + i \sin \left(\frac{-5\pi}{6} \times 6 \right) \right] \quad (\text{De Moivre})$$

$$= 2^6 [\cos(-5\pi) + i \sin(-5\pi)]$$

$$= 64(-1 + 0i)$$

$$= -64$$

11. Complex Numbers, EXT2 N1 2016 HSC 11a

i. $z = \sqrt{3} - i$

$$|z| = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$\begin{aligned}\therefore z = \sqrt{3} - i &= 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) \\ &= 2 \left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) \right) \\ &= 2\text{cis}\left(-\frac{\pi}{6}\right)\end{aligned}$$

ii. $z^6 = 2^6 \left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) \right)^6$

$$= 64 \text{cis}(-\pi) \quad (\text{by De Moivre})$$

$$= -64$$

$\therefore z^6$ is real.

iii. $z^n = 2^n \left(\cos\left(-\frac{n\pi}{6}\right) + i\sin\left(-\frac{n\pi}{6}\right) \right)$

z^n is purely imaginary when:

$$\cos\left(-\frac{n\pi}{6}\right) = 0$$

Or more generally,

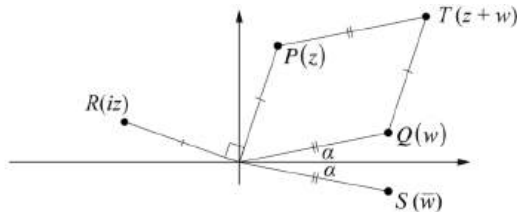
$$\frac{n\pi}{6} = \frac{\pi}{2} + k\pi$$

$$n = 6k + 3, \quad (k \in \mathbb{Z})$$

$$\therefore n = 3, 9, 15, \dots \quad (n > 0)$$

12. Complex Numbers, EXT2 N1 2009 HSC 2c

i, ii, and iii.



13. Complex Numbers, EXT2 N1 2014 HSC 11a

i. $z + w = -2 - 2i + 3 + i$

$$= 1 - i$$

$$|z + w| = \sqrt{2}$$

$$\arg(z + w) = -\frac{\pi}{4}$$

$$\therefore z + w = \sqrt{2} \text{cis}\left(-\frac{\pi}{4}\right)$$

ii. $\frac{z}{w} = \frac{-2 - 2i}{3 + i}$

$$= \frac{(-2 - 2i)(3 - i)}{(3 + i)(3 - i)}$$

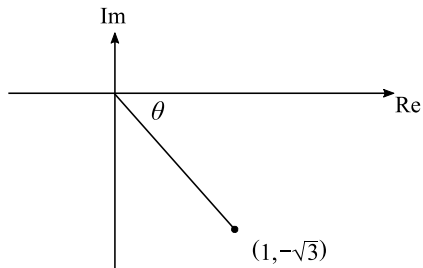
$$= \frac{-6 - 6i + 2i - 2}{9 + 1}$$

$$= -\frac{8}{10} - \frac{4}{10}i$$

$$= -\frac{4}{5} - \frac{2}{5}i$$

14. Complex Numbers, EXT2 N1 2005 HSC 2b

i. $\beta = 1 - i\sqrt{3}$



$$|\beta| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\tan\theta = \frac{\sqrt{3}}{1} = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

$$\arg(\beta) = -\frac{\pi}{3}$$

$$\therefore \beta = 2 \operatorname{cis}\left(-\frac{\pi}{3}\right)$$

ii. $\beta^5 = 2^5 \operatorname{cis}\left(-\frac{\pi}{3} \times 5\right)$

$$= 32 \operatorname{cis}\left(-\frac{5\pi}{3} + 2\pi\right)$$

$$= 32 \operatorname{cis}\left(\frac{\pi}{3}\right)$$

iii. $\beta^5 = 32\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right)$

$$= 32\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= 16 + i16\sqrt{3}$$

15. Complex Numbers, EXT2 N1 2019 HSC 11e

i. $|z| = -1 + i\sqrt{3}$

$$= \sqrt{(-1)^2 + (\sqrt{3})^2}$$

$$= 2$$

$$\tan\theta = -\sqrt{3}$$

$$\arg(z) = \frac{2\pi}{3}$$

$$\therefore z = 2\operatorname{cis}\frac{2\pi}{3}$$

ii. $z^3 = 2^3\left[\cos\left(3 \times \frac{2\pi}{3}\right) + i\sin\left(3 \times \frac{2\pi}{3}\right)\right]$ (by De Moivre)

$$= 8(\cos 2\pi + i\sin 2\pi)$$

$$= 8(1 + 0i)$$

$$= 8 + 0i$$

16. Complex Numbers, EXT2 N1 2004 HSC 2b

$$\begin{aligned} \text{i. } \frac{\alpha}{\beta} &= \frac{1+i\sqrt{3}}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{(1+i\sqrt{3})(1-i)}{1^2-i^2} \\ &= \frac{1-i+i\sqrt{3}-i^2\sqrt{3}}{2} \\ &= \frac{1+\sqrt{3}}{2} + i\left(\frac{\sqrt{3}-1}{2}\right) \end{aligned}$$

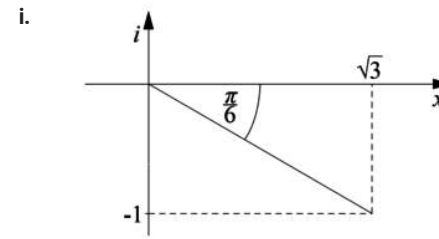
$$\begin{aligned} \text{ii. } \alpha &= 1+i\sqrt{3} \\ |\alpha| &= \sqrt{1^2 + (\sqrt{3})^2} = 2 \\ \arg(\alpha) &= \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3} \\ \therefore \alpha &= 2\text{cis}\left(\frac{\pi}{3}\right) \end{aligned}$$

$$\begin{aligned} \text{iii. } \beta &= \sqrt{2}\text{cis}\left(\frac{\pi}{4}\right) \\ \frac{\alpha}{\beta} &= \frac{2}{\sqrt{2}} \text{cis}\left(\frac{\pi}{3}-\frac{\pi}{4}\right) \\ &= \sqrt{2}\text{cis}\left(\frac{\pi}{12}\right) \end{aligned}$$

iv. Equating imaginary parts of i and ii:

$$\begin{aligned} \sqrt{2} \sin\left(\frac{\pi}{12}\right) &= \frac{\sqrt{3}-1}{2} \\ \sin\left(\frac{\pi}{12}\right) &= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{6}-\sqrt{2}}{4} \end{aligned}$$

17. Complex Numbers, EXT2 N1 2006 HSC 2b



$$\begin{aligned} |\sqrt{3}-i| &= \sqrt{(\sqrt{3})^2 + 1^2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \theta &= \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) \\ &= -\frac{\pi}{6} \end{aligned}$$

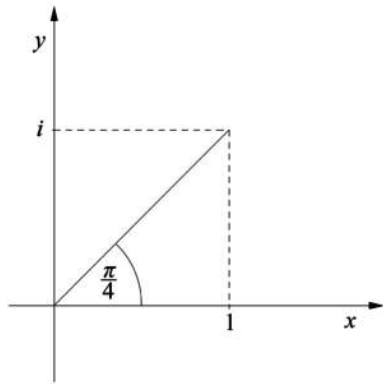
$$\therefore \sqrt{3}-i = 2\text{cis}\left(-\frac{\pi}{6}\right)$$

$$\begin{aligned} \text{ii. } (\sqrt{3}-i)^7 &= 2^7\text{cis}\left(-\frac{7\pi}{6}\right) \quad (\text{De Moivre}) \\ &= 128\text{cis}\left(\frac{5\pi}{6}\right) \end{aligned}$$

$$\begin{aligned} \text{iii. } (\sqrt{3}-i)^7 &= 128\left(\cos \frac{5\pi}{6} + i\sin \frac{5\pi}{6}\right) \\ &= 128\left(-\frac{\sqrt{3}}{2} + \frac{i}{2}\right) \\ &= -64\sqrt{3} + 64i \end{aligned}$$

18. Complex Numbers, EXT2 N1 2007 HSC 2b

i.



$$|1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\arg(1 + i) = \frac{\pi}{4}$$

$$\therefore 1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

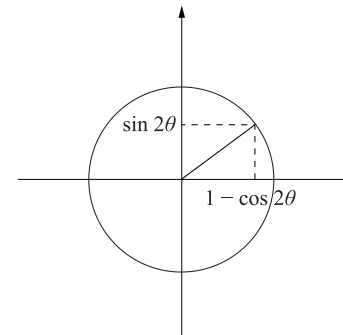
$$\begin{aligned} \text{ii. } (1 + i)^{17} &= \left(\sqrt{2} \right)^{17} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^{17} \\ &= 2^8 \sqrt{2} \left(\cos \frac{17\pi}{4} + i \sin \frac{17\pi}{4} \right) \quad (\text{De Moivre}) \\ &= 2^8 \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\ &= 2^8 \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) \\ &= 2^8 (1 + i) \\ &= 256 + 256i \end{aligned}$$

19. Complex Numbers, EXT2 N1 2018 HSC 13b

i. $z = 1 - \cos 2\theta + i \sin 2\theta$

$$\begin{aligned} |z| &= \sqrt{(1 - \cos 2\theta)^2 + \sin^2 2\theta} \\ &= \sqrt{1 - 2\cos 2\theta + \cos^2 2\theta + \sin^2 2\theta} \\ &= \sqrt{2 - 2\cos 2\theta} \\ &= \sqrt{2(1 - \cos 2\theta)} \\ &= \sqrt{2(2\sin^2 \theta)} \\ &= 2\sin \theta \quad \dots \text{ as required} \end{aligned}$$

ii. $z = 1 - \cos 2\theta + i \sin 2\theta$



$$\begin{aligned} \arg(z) &= \tan^{-1} \left(\frac{\sin 2\theta}{1 - \cos 2\theta} \right) \\ &= \tan^{-1} \left(\frac{2\sin \theta \cos \theta}{2\sin^2 \theta} \right) \\ &= \tan^{-1}(\cot \theta) \\ &= \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \theta \right) \right) \\ &= \frac{\pi}{2} - \theta \end{aligned}$$

$$\left(\text{Since } 0 < \theta < \pi \Rightarrow -\frac{\pi}{2} \leq \frac{\pi}{2} - \theta < \frac{\pi}{2} \right)$$

20. Complex Numbers, EXT2 N1 2015 HSC 11b

$$\text{i. } |z| = \sqrt{(-\sqrt{3})^2 + 1^2}$$

$$= 2$$

$$\text{ii. } \arg(z) = \tan^{-1}(1 - \sqrt{3})$$

$$= \pi - \frac{\pi}{6}$$

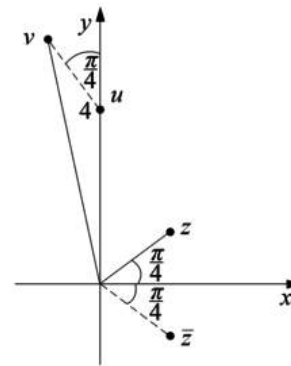
$$= \frac{5\pi}{6}$$

$$\text{iii. } \arg\left(\frac{z}{w}\right) = \arg z - \arg w$$

$$= \frac{5\pi}{6} - \frac{\pi}{7}$$

$$= \frac{29\pi}{42}$$

21. Complex Numbers, EXT2 N1 2015 HSC 12a



$$\text{i. } z = 2\text{cis}\frac{\pi}{4}$$

$$= \sqrt{2}(1 + i)$$

$$\text{ii. } u = z^2$$

$$= 4\text{cis}\frac{\pi}{2}$$

$$= 4i$$

$$\text{iii. } v = z^2 - \bar{z}$$

$$= 4i - \sqrt{2}(1 - i)$$

$$= -\sqrt{2} + (4 + \sqrt{2})i$$

COMMENT: 12a(iii) had the lowest mean mark (55%) of any part within Q12 and deserves attention.

22. Complex Numbers, EXT2 N1 2008 HSC 2b

$$\begin{aligned} \text{i. } \frac{1+i\sqrt{3}}{1+i} \times \frac{1-i}{1-i} &= \frac{(1+i\sqrt{3})(1-i)}{1-i^2} \\ &= \frac{1-i+i\sqrt{3}-\sqrt{3}i^2}{2} \\ &= \frac{1+\sqrt{3}}{2} - i\left(\frac{1-\sqrt{3}}{2}\right) \end{aligned}$$

$$\text{ii. } z_1 = 1 + i\sqrt{3}$$

$$|z_1| = \sqrt{1 + (\sqrt{3})^2} = 2$$

$$\arg(z_1) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$z_1 = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$z_2 = 1 + i$$

$$|z_2| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\arg(z_2) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$z_2 = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

$$\begin{aligned} \frac{1+i\sqrt{3}}{1+i} &= \frac{z_1}{z_2} \\ &= \frac{2}{\sqrt{2}}\left(\cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{3}-\frac{\pi}{4}\right)\right) \\ &= \sqrt{2}\left(\cos\left(\frac{\pi}{12}\right) + i\sin\left(\frac{\pi}{12}\right)\right) \end{aligned}$$

iii. Equating real parts of i and ii:

$$\sqrt{2}\cos\left(\frac{\pi}{12}\right) = \frac{1+\sqrt{3}}{2}$$

$$\begin{aligned} \cos\left(\frac{\pi}{12}\right) &= \frac{1+\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2}+\sqrt{6}}{4} \end{aligned}$$

$$\text{iv. } \left(\frac{1+i\sqrt{2}}{1+i}\right)^{12} = (\sqrt{2})^{12}\left(\cos\left(\frac{\pi}{12} \times 12\right) + i\sin\left(\frac{\pi}{12} \times 12\right)\right)$$

$$= 64(\cos\pi + i\sin\pi)$$

$$= -64$$